

Alex Rop

Geospatial Engineer · Founder, ROXR Earth

Nairobi, Kenya · roplex.github.io/MyWebsite · linkedin.com/in/alex-rop44

PROFILE

Geospatial Engineer using satellite intelligence to support the people and places most exposed to environmental risk. I combine Earth Observation, GIS and machine learning in reproducible workflows that turn imagery into practical information for drought monitoring, climate risk and environmental decision-making across Africa. Based in Nairobi, working with international partners remotely and with communities and stakeholders in the field.

EXPERIENCE

Ventures Incubatee / Fellow

Sep 2026 – Present

United Nations Institute for Training and Research (UNITAR) **REMOTE**

- Participant in Phase I of the Empower Africa Programme (TICAD9), UNITAR's Japan-Africa public-private partnership programme on entrepreneurship, AI and digital transformation, business model development and investment readiness, with ROXR Earth as my live project.

Founder

Jun 2026 – Present

ROXR Earth Limited · Nairobi

- Designed and operate a six-layer Python pipeline behind the ROXR Climate Intelligence Index (RCII), fusing MODIS vegetation and land surface temperature imagery with CHIRPS rainfall into a drought index normalised by county and season.
- Trained three Random Forest models on 26 years of history to forecast drought severity 30, 60 and 90 days ahead, validated against Kenya's 2011, 2017 and 2022 droughts (2.2–2.7× better than chance).
- Produce monthly PDF intelligence briefs, an interactive Folium dashboard and JSON/CSV/GeoJSON data feeds for agricultural insurers, development finance institutions and humanitarian agencies.

Geospatial & Stakeholder Engagement Consultant

Aug 2025 – Present

Futures Inc. (Japan) · Lake Victoria, Kenya **HYBRID**

- GIS and remote sensing for spatial site selection and land use assessment on the Yamaha Carbon Credit Project, a nature-based carbon project, coordinating across international stakeholders with field visits to meet local stakeholders.

Geospatial Data Analyst Consultant

Jan 2025 – Feb 2026

Fairtrade Africa · Eastern and Central Africa **HYBRID**

- Led land cover change detection and deforestation risk monitoring across 63+ Producer Organizations in six East and Central African countries, using OpenStreetMap, satellite imagery and thematic maps.
- Managed validation and quality assurance of monitoring data used in EU Deforestation Regulation (EUDR) compliance reporting.
- Developed and delivered geospatial data interpretation training to field teams, virtually and in person, improving data collection quality and spatial literacy among non-technical staff.

Geospatial Research Assistant & SDL Intern

Aug 2024 – May 2025

Harvard University, Center for Geographic Analysis **REMOTE**

- Developed automated Python and KNIME workflows integrating MODIS, Landsat, CHIRPS and soil moisture data across a 23-year time series for multi-source drought index monitoring, including drought analysis in India. Supervised by Devika Jain, AI and Data Science Manager.

Internship Trainee

Jan 2024 – Mar 2024

Esri Eastern Africa · Nairobi

- Built a disaster risk prototype for Kenya's ASAL counties combining suitability modelling, Random Forest drought forecasting and ArcGIS Dashboard visualization for government and humanitarian stakeholders.

Freelance Geospatial Engineer

Feb 2023 – Nov 2023

Uasin Gishu County · Eldoret

- GIS consultancy including work with the Clerk of Works at Kapseret CDF, supporting spatial data management and mapping for county-level infrastructure and development planning.

Attaché

Jan 2022 – Apr 2022

Ministry of Lands and Physical Planning · Eldoret

- Led data collation for the Nine Farms Survey Project, a multi-agency cadastral survey with the Uasin Gishu County Government and the national Ministry of Lands; took part in land subdivision surveys, land dispute resolution fieldwork, topographic surveying and boundary verification.

PUBLICATIONS & PRESENTATIONS

Jain, D. & Rop, A. (forthcoming). *Harmonizing Earth Observation Data for Enhanced Spatial Analysis*. Book chapter, Elsevier. A KNIME-based workflow for multi-source Earth Observation data integration, with a case study on surface urban heat islands in Phoenix, Arizona.

Automated CHIRPS Rainfall Data Extraction for Multi-Source Drought Index Monitoring. Team presentation, Harvard CGA Conference, 2025.

EDUCATION

B.Eng, Geospatial Engineering · Second Class Honours (Upper Division)

2016 – 2021

The Technical University of Kenya

- Final-year project: mapping changes in built-up areas, with Mombasa Road, Nairobi as the case study.

TECHNICAL SKILLS

Programming	Python (pandas, NumPy, SciPy, scikit-learn, GeoPandas, Rasterio, GDAL, matplotlib, Folium), R, JavaScript for Google Earth Engine, KNIME
Earth Observation	Google Earth Engine, MODIS, Landsat, Sentinel-2, CHIRPS; anomaly detection, time-series analysis, land cover classification
Machine learning	Random Forest classification, time-series cross-validation, feature engineering, class imbalance handling, feature importance analysis
GIS & spatial data	QGIS, ArcGIS Pro, PostGIS/PostgreSQL, coordinate reference systems, zonal statistics, spatial data management
Data formats	GeoTIFF, NetCDF, Shapefile, GeoJSON, CSV
Reproducibility	Git/GitHub, pipeline run logging, documented methodology, reproducible workflow design

TRAINING & CERTIFICATIONS

- Spatial Data Science: The New Frontier in Analytics · Esri, 2024
- GIS for Climate Action · Esri, 2024
- AiCE, Career Essentials · ALX, 2024
- Machine Learning for Everyone · DataCamp, 2021
- AI Fundamentals: Core Concepts and Principles · edX
- AI in Society: Ethics, Leadership, and the Road Ahead · edX
- From Space to Screen

LANGUAGES

English (full professional) · Kiswahili (native) · Kalenjin (professional working)

REFERENCES

Available on request.